Master Project Bank

Cloud Computing and Cross-Course Project Options for CS-Only, AI-Only, and Mixed Teams

Purpose	Provide one approval-ready project bank that unifies the CS-only, AI-only, and mixed-team shortlists into a single reference document.
Design principle	Every approved project should have **a controlled scope**, **a clear course contribution**, and **a coherent cloud deployment story**.
Prepared for	Cloud Computing Spring 2026 and cross-course coordination with Software Engineering 2, Network Security, Computer Graphics, Compiler, Natural Language Understanding, Speech Processing, Evolutionary Algorithms, and Computer Vision.

How to use this document

This bank is organized by **team type** so supervisors can approve projects using the same structure across the semester. *Each project entry includes a one-line summary, course mapping, recommended scope, and approval notes.* The goal is to help teams choose a topic that is ambitious enough to be meaningful but still narrow enough to finish well.

Approval principles

- The project must fit the team profile. Product-first topics suit CS-only teams; model-first topics suit AI-only teams; layered systems suit mixed teams.
- The project must have a clear primary academic contribution for every participating course. Teams should not rely on one course doing all the heavy lifting.
- **The cloud contribution should be meaningful.** A project should have a credible AWS deployment, storage, monitoring, security, or architecture story rather than a token hosting step.
- Scope discipline matters more than feature count. A smaller project with a strong architecture and clean evaluation is preferable to an oversized system with weak execution.

Project categories at a glance

Team type	Primary project style	Typical course anchors	Count
CS-only	Product-first	Software Engineering 2, Network Security, Computer Graphics, Compiler, Cloud	5
AI-only	Model-first / algorithm-first	NLU, Speech Processing, Evolutionary Algorithms, Computer Vision, Cloud	6
Mixed	Layered systems	SE2 + AI course(s) + Cloud	7

Category A | CS-only teams

These projects are best approved when the team's main strength is software engineering, APIs, roles and authorization, Docker, database-backed workflows, and product delivery. Cloud should be treated as the deployment and operations layer rather than as a separate unrelated project.

Approval shortlist

#	Project title	Core style	Course mapping	Approval
1	Secure Campus Services Portal	Product-first	Software Engineering 2 + Network Security + Cloud Computing	Standard
2	Smart Clinic and Appointment Management System	Product-first	Software Engineering 2 + Network Security + Cloud Computing	Standard
3	Logistics and Warehouse Operations Platform	Product-first	Software Engineering 2 + Cloud Computing	Standard
4	Unity Game Backend and Player Services Platform	Product-first	Computer Graphics + Software Engineering 2 + Cloud Computing	Conditional
5	Compiler-as-a-Service Learning Platform	Product-first	Compiler Project + Software Engineering 2 + Cloud Computing	Standard

Detailed project entries

1. Secure Campus Services Portal

Project summary	A university-facing workflow system for student requests, appointments, announcements, document handling, and approvals.
Course mapping	Software Engineering 2 + Network Security + Cloud Computing
Recommended scope	Build 4–5 production-style modules: authentication/roles, request management, appointment booking, document handling, and notifications. Deploy the full stack as containerized services with cloud storage for uploaded files.
Approval notes	Recommended as a default CS option. Strong fit for Spring Boot microservices, JWT-based authorization, UML/SRS deliverables, and cloud deployment. Avoid adding too many unrelated campus subsystems.

2. Smart Clinic and Appointment Management System

Project summary	A role-based healthcare operations platform for patient records, doctor schedules, appointments, visit summaries, and notifications.
Course mapping	Software Engineering 2 + Network Security + Cloud Computing

Recommended scope	Keep the medical scope operational rather than diagnostic: registration, patient records, schedules, appointments, and visit documentation. Use private database access patterns and secure API authorization.
Approval notes	Recommended when the team wants a realistic enterprise-style system. Approved only if the team avoids attempting advanced diagnosis or AI features that would broaden scope unnecessarily.

3. Logistics and Warehouse Operations Platform

Project summary	A business operations system for inventory, warehouse movements, orders, shipments, and staff dashboards.
Course mapping	Software Engineering 2 + Cloud Computing
Recommended scope	Implement 4–6 modules such as auth, inventory, orders, warehouse movement, shipment tracking, and reporting. Add a cloud deployment story around observability, storage, and scaling assumptions.
Approval notes	Recommended for teams that want clean microservice boundaries and strong dashboard/reporting workflows. Keep route optimization or ML recommendations out unless separately approved.

4. Unity Game Backend and Player Services Platform

Project summary	A backend platform for a Unity game that provides player identity, profiles, leaderboards, saved progress, and session analytics.
Course mapping	Computer Graphics + Software Engineering 2 + Cloud Computing
Recommended scope	Treat the Unity game as the client and build the cloud-native service layer only: auth, profiles, leaderboard, cloud save, and admin dashboard. Exclude full real-time multiplayer infrastructure unless the game already demands it and the scope is controlled.
Approval notes	Conditionally recommended for teams already committed to a Unity game. Approval depends on keeping the backend scope focused and not turning the project into a full game-engine/networking research effort.

5. Compiler-as-a-Service Learning Platform

Project summary	A web platform that exposes compiler phases as services for code submission, tokenization, parsing, error reporting, and history.
Course mapping	Compiler Project + Software Engineering 2 + Cloud Computing
Recommended scope	Restrict the academic core to the first two compiler phases and wrap them in a proper service workflow: auth, submissions, lexer/parser results, history, and admin review.
Approval notes	Recommended for teams that want a strong bridge to the compiler course. Approval depends on limiting scope to early compiler phases and emphasizing the service platform around them.

Category B | AI-only teams

These projects are best approved when the team's main strength is modeling, fine-tuning, optimization, experimentation, and evaluation. The system shell can remain lightweight, but the AI contribution must be academically substantive and measurable.

Approval shortlist

#	Project title	Core style	Course mapping	Approval
1	Arabic Document Intelligence Platform	Model-first	Natural Language Understanding + Cloud Computing	Standard
2	Arabic Speech-to-Text Improvement Platform	Model-first	Speech Processing + Cloud Computing	Standard
3	Cloud Resource Allocation Optimizer	Model-first	Evolutionary Algorithms + Cloud Computing	Standard
4	Vision Feature Selection / Neural Evolution Platform	Model-first	Computer Vision + Evolutionary Algorithms + Cloud Computing	Standard
5	Adaptive Recommendation or Search Optimization Engine	Model-first	Natural Language Understanding + Evolutionary Algorithms + Cloud Computing	Conditional
6	Resume / Invoice / Clinical Information Extraction System	Model-first	Natural Language Understanding + Cloud Computing	Standard

Detailed project entries

Cloud Computing plays a pivotal role across all projects in this category by providing robust infrastructure and scalability. Its contributions include enabling deployment architecture, facilitating containerization for modular and efficient application management, and supporting storage of experiment runs to ensure reproducibility and data tracking. Additionally, it offers comprehensive monitoring and logging capabilities for system reliability and performance analysis. Projects may also leverage cloud dashboards for interactive simulation and management, enhancing user engagement and operational transparency.

6. Arabic Document Intelligence Platform

Project summary	A document-focused AI system for Arabic summarization, structured extraction, classification, or document question answering.
Course mapping	Natural Language Understanding + Cloud Computing
Recommended scope	Choose one clear task and one base model. Fine-tune the same selected model with LoRA, compare base vs fine-tuned performance, and demonstrate structured outputs with a lightweight demo interface.
Approval notes	Recommended as a flagship AI option. Approval depends on narrowing the task to one well-defined document problem and avoiding a vague all-in-one assistant.

7. Arabic Speech-to-Text Improvement Platform

Project summary	A research-driven speech system for Arabic transcription improvement in settings such as lecture audio, noisy speech, or command-style input.
Course mapping	Speech Processing + Cloud Computing
Recommended scope	Start from a baseline recognizer or pipeline, implement a measurable improvement, and evaluate results quantitatively. Use cloud services for audio storage, queued processing, and demo hosting.
Approval notes	Recommended when the team has a strong speech focus and can define a concrete improvement objective. **Not approved if the project is only a thin wrapper around an off-the-shelf model.**

8. Cloud Resource Allocation Optimizer

Project summary	A simulation and optimization project that allocates cloud workloads to resources using evolutionary methods.
Course mapping	Evolutionary Algorithms + Cloud Computing
Recommended scope	Formalize a cloud scheduling/allocation problem, define fitness objectives such as cost and response time, and implement GA, ACO, or a hybrid GA–ACO approach with repeated experiments and visual reporting.
Approval notes	Highly recommended for optimization-oriented teams. Approval requires rigorous experimentation, multiple settings, and clear visualization of results rather than a superficial UI.

9. Vision Feature Selection / Neural Evolution Platform

Project summary	An image-analysis platform that uses evolutionary methods for feature selection, parameter search, or neural evolution in a vision task.
Course mapping	Computer Vision + Evolutionary Algorithms + Cloud Computing
Recommended scope	Choose one vision task, build a baseline pipeline, then optimize a meaningful component using GA/DE or a related EA approach. Provide comparative experiments and a results dashboard.
Approval notes	Recommended for advanced AI teams with strong experimentation capacity. Approval depends on selecting a manageable dataset and a narrow optimization target.

10. Adaptive Recommendation or Search Optimization Engine

Project summary	A recommendation or search system that combines semantic retrieval with adaptive ranking or recommendation optimization.
Course mapping	Natural Language Understanding + Evolutionary Algorithms + Cloud Computing
Recommended scope	Build a baseline search/recommendation system, then add either natural-language query understanding, evolutionary reranking, or both. Keep the product domain small: books, courses, or products.
Approval notes	Conditionally recommended for stronger teams. Approval depends on having a clear baseline and a small domain so evaluation remains credible.

11. Resume / Invoice / Clinical Information Extraction System

Project summary	A structured-output AI system that extracts key fields from resumes, invoices, or clinical notes.
Course mapping	Natural Language Understanding + Cloud Computing
Recommended scope	Pick one domain only, define the output schema clearly, fine-tune with **LoRA**, and evaluate extraction quality before and after fine-tuning. Package the model behind a simple interface or inference service.
Approval notes	Recommended because the tasks are objective and easy to benchmark. Approval depends on staying focused on one document type and one extraction objective.

Category C | Mixed teams

These projects work best when the team naturally splits into a CS side, an AI side, and a cloud/deployment side. The final system should feel integrated rather than like three disconnected mini-projects.

Approval shortlist

#	Project title	Core style	Course mapping	Approval
1	Resume / Candidate Intelligence Platform	Layered system	Software Engineering 2 + Natural Language Understanding + Cloud Computing	High
2	Invoice / Document Intelligence System	Layered system	Software Engineering 2 + Natural Language Understanding + Cloud Computing	High
3	Arabic Voice-Enabled Student Assistant	Layered system	Speech Processing + Software Engineering 2 + Cloud Computing	Standard
4	Vision Analysis and Reporting Platform	Layered system	Computer Vision + Evolutionary Algorithms + Software Engineering 2 + Cloud Computing	Standard
5	Intelligent Product Search and Recommendation Platform	Layered system	Natural Language Understanding + Evolutionary Algorithms + Software Engineering 2 + Cloud Computing	High
6	Cloud Resource Scheduling and Operations Dashboard	Layered system	Evolutionary Algorithms + Software Engineering 2 + Cloud Computing	Standard
7	Automated Essay Feedback Platform	Layered system	Natural Language Understanding + Software Engineering 2 + Cloud Computing	Standard

Detailed project entries

12. Resume / Candidate Intelligence Platform

Project summary	A recruitment platform that combines resume upload, structured candidate profiling, recruiter dashboards, and searchable candidate workflows.
Course mapping	Software Engineering 2 + Natural Language Understanding + Cloud Computing
Recommended scope	**CS students** build the recruitment shell and workflows; **AI students** fine-tune resume parsing/candidate profiling; cloud students deploy document storage, services, and monitoring.
Approval notes	Top mixed-team recommendation. Strongly approved because the split between product, AI extraction, and cloud infrastructure is clean and fair.

13. Invoice / Document Intelligence System

Project summary	A business-document platform that ingests invoices or long-form documents, extracts key information, and supports human review and searchable records.
Course mapping	Software Engineering 2 + Natural Language Understanding + Cloud Computing
Recommended scope	**CS students** own workflow and dashboards; **AI students** own extraction and structured output; cloud students own storage, deployment, and monitoring. **Keep to one document type or one document family.**
Approval notes	Strongly recommended. Especially suitable when the teaching team wants an easily assessable project with clear structured outputs and strong demo value.

14. Arabic Voice-Enabled Student Assistant

Project summary	An education-oriented assistant that accepts spoken Arabic input, performs transcription or voice understanding, and routes requests through a role-based service platform.
Course mapping	Speech Processing + Software Engineering 2 + Cloud Computing
Recommended scope	**AI students** build the voice pipeline; **CS students** build the role-based shell, history, and service orchestration; cloud students manage audio storage, asynchronous processing, and observability.
Approval notes	Recommended for teams with a clear speech focus. Approval depends on choosing a limited assistant scope such as academic FAQs, service requests, or timetable queries.

15. Vision Analysis and Reporting Platform

Project summary	A reporting platform for image analysis tasks such as screening, inspection, or feature-based comparison, with optional optimization support.
Course mapping	Computer Vision + Evolutionary Algorithms + Software Engineering 2 + Cloud Computing
Recommended scope	**AI students** build the vision and optional optimization core; **CS students** build upload, reporting, and review workflows; cloud students host storage and compute services.
Approval notes	Recommended when the team already has strong vision-course alignment. Approval depends on picking one narrow image task and not trying to cover multiple problem families.

16. Intelligent Product Search and Recommendation Platform

Project summary	A smart catalog platform that supports natural-language search and optimized recommendation or reranking.
Course mapping	Natural Language Understanding + Evolutionary Algorithms + Software Engineering 2 + Cloud Computing
Recommended scope	**Keep the domain small.** **CS students** build catalog/workflow modules; **AI students** handle query understanding and/or ranking optimization; cloud students deploy and monitor the stack.
Approval notes	Top-tier mixed-team option for advanced groups. **Approval depends on a very clear baseline and disciplined scope because this can expand quickly.**

17. Cloud Resource Scheduling and Operations Dashboard

Project summary	A dashboard-driven cloud operations platform that visualizes workload scheduling while an optimization engine allocates resources behind the scenes.
Course mapping	Evolutionary Algorithms + Software Engineering 2 + Cloud Computing
Recommended scope	**AI students** own the optimization engine and experiments; **CS students** build the dashboard and reports; cloud students justify architecture, observability, and operations decisions.
Approval notes	**Recommended for technically mature teams** that prefer simulation and operations over user-facing business workflows.

18. Automated Essay Feedback Platform

Project summary	An education platform where students submit essays and receive model-generated, structured feedback while instructors review results and manage submissions.
Course mapping	Natural Language Understanding + Software Engineering 2 + Cloud Computing
Recommended scope	**CS students** build roles and submission workflows; **AI students** fine-tune and evaluate the feedback model; cloud students deploy, store, and monitor the service.
Approval notes	Recommended in education-oriented settings. **Approval depends on defining a clear feedback schema and avoiding unconstrained open-ended generation.**